Complete Cloud Computing Syllabus For BCA

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Module 1: Semester 1

You get to learn the following cloud computing topics typically in BCA.

- **Algebra, Calculus, and Trigonometry:** Learn about matrices, theory of equations, differential calculus, integral calculus, and trigonometry.
- **Programming in C:** Get an overview of programming, learn the fundamentals of C programming, advanced programming techniques, dynamic data structures in C, and many other additional features.
- **Introduction to Linux:** Introduce yourself to Linux, learn the Unix file system, Unix process management, VI editor, and system administration.
- Laboratory works on C programming and Linux programming.

Module 2: Semester II

- **Probability and Statistics:** Learn about random variables, distribution functions, moments, moment-generating functions, two-dimensional random variables, descriptive statistics, and correlation and curve fitting.
- **Fundamentals of storage management:** Introduction to information storage and management, learning storage system management, raid and storage networking technologies, backup and recovery, and local and remote replication techniques.
- **Operating systems:** Introduction to O.S., process management, memory management, file-system and disk management, and security systems.
- **OOPS with C++:** Introduction to object-oriented programming with C++, learn what a class is, operator overloading, what a pointer is, and templates and exception handling.

- **Data Structure with C:** Introduction to data structures in C, searching and sorting, stack and queue, linked list, and tree graphs with their applications.
- Lab Work on OOPS with C++ and data structures.

Module 3: Semester III

- **Information security fundamentals:** Introduction to information technology, what is the need for information technology, risk management methods, network infrastructure security and connectivity, and information asset classification.
- **Software engineering:** Learn software products, processes, requirements, analysis, design concepts, principles, testing, and project management.
- **RDBMS:** Introduction to Relational Database management Systems, relational models, SQL views, index, triggers, database design, and transactions.
- **Computer Networks:** Learn the fundamentals of networking, basics of network devices, network transport and application layers, WAN technology, and network operating systems and troubleshooting networks.
- **JAVA Programming:** Introduction to programming in JAVA, learn about the classes, packages, multithreaded programming, and JAVA database connectivity
- Lab Work on RDBMS and Programming in Java.

Module 4: Semester IV

You get to study the following subjects as part of module 4.

- **Server operating system:** Learn installing and configuring servers, configure local storages, configure file and share access, and configure print, document services, and servers for remote management.
- **Ethical hacking fundamentals:** Learn what ethical hacking is, how to do ethical hacking using attacks and exploits, web and network hacking, report writing and mitigation, and the legalities of ethical hacking.
- **Cryptography fundamentals:** Introduction to cryptography, different types of algorithms, key management, cryptography applications, and use of cryptography in user authentication.
- **Introduction to cloud technology:** Learn about different cloud computing companies and migrating to the Cloud, management of cloud cost and how you

should select a cloud provider, how the Cloud is governed, and what are the risks and benefits of cloud computing.

- **Fundamentals of the data center:** Get an overview of data centers, requirements of data centers, design of data centers, introduction to server farms, and business continuity and disaster recovery fundamentals.
- Lab work on ethical hacking and server operating systems.

Module 5: Semester V

You study the following topics as a part of module 5.

- **Computer Forensics:** Learn about computer forensics, storage devices, data recovery methods, forensic techniques, and cyber laws.
- **Virtualization and cloud security:** Introduction to virtualization in the Cloud, cloud security, cloud trust protocol, transparency, cloud data security, and the legalities of cloud security.
- **I.T. governance, risks, and information security management:** Know what I.T. governance is, what the different information systems strategies are, risk management programs, and information security management
- **Introduction to cloud computing solutions:** It includes infrastructure solutions on the Cloud, getting started with Azure, Azure storage, and Azure networking. Azure active directory, Azure database, and Azure websites.
- **Principles of virtualization:** Learn the basics of virtualization, deployment and managing an enterprise desktop virtualization environment, deployment and managing a presentation virtualization environment, accessing published applications, and understanding virtualization software.
- The lab works on computer forensics and principles of virtualization.

Module 6: Semester VI

- Mobile, wireless, and VOIP security: Introduction to wireless technology, wireless security, voice over internet protocol security, mobile forensics, data extraction, and mobile application development security.

Modules	Semesters	Subjects
1	I	• Algebra, Calculus, and Trigonometry • Programming in C • Introduction to Linux • Laboratory works on C programming and Linux programming
2	II	• Probability and Statistic • Fundamentals of storage management • Operating system • OOPS with C++ • Data Structure with C • Lab Work on OOPS with C++ and data structures.
3	III	• Information security fundamentals • Software engineering • RDBMS • Computer Networks • JAVA Programming • Lab Work on RDBMS and Programming in Java
4	IV	• Server operating system • Ethical hacking fundamental

		• Cryptography fundamentals • Introduction to cloud technology • Fundamentals of the data center • Lab Work on ethical hacking and server operating systems
5	V	• Computer Forensics • Virtualization and cloud security • IT governance, risks, and information security managemen • Introduction to cloud computing solutions • Principles of virtualization • The lab works on computer forensics and principles of virtualization
6	VI	• Mobile, wireless, and VOIP security • Grand project and viva.